# FLICS 2026 Presentation – Possible Questions & Answers

This document contains possible questions that may be asked during the FLICS 2026 presentation and discussion session, along with professional sample answers for preparation.

## Q1. Why did you choose Federated Learning instead of centralized learning?

Centralized learning requires sharing sensitive patient data across hospitals, which creates privacy and regulatory concerns such as HIPAA and GDPR compliance. Federated Learning allows collaborative model training while keeping patient data locally at each hospital, making it more suitable for real-world healthcare environments.

## Q2. Why did you use the OPTN dataset?

The OPTN dataset is one of the largest and most comprehensive kidney transplant registries, containing real-world transplant data from multiple centers across the United States. It provides diverse clinical, donor, and transplant-related features, making it suitable for evaluating federated learning in realistic healthcare scenarios.

## Q3. Why did you select only 5 transplant centers?

We selected the top 5 transplant centers with the highest number of transplant records to ensure sufficient data distribution across clients while maintaining computational feasibility for the initial study. It also helped simulate a realistic multi-center federated environment.

## Q4. What is Non-IID data and why is it important here?

Non-IID means the data distribution is different across hospitals. Patient demographics, donor profiles, and treatment practices vary between centers. This reflects real-world healthcare systems, and handling Non-IID data is one of the biggest challenges in Federated Learning.

## Q5. Why did FedAvg and FedProx perform better than SCAFFOLD?

FedAvg and FedProx handled the medical data heterogeneity more effectively in our setup. SCAFFOLD uses control variates to reduce client drift, but in highly heterogeneous healthcare data, this sometimes introduced instability during optimization.

## Q6. Why did you choose an MLP instead of more advanced models?

We selected MLP because it provides a strong baseline for tabular clinical data and is computationally efficient in federated settings. This work mainly focuses on evaluating federated learning feasibility in kidney transplant prediction.

## Q7. Why did you use Balanced Accuracy instead of normal accuracy?

The dataset is imbalanced because graft failure cases are much fewer than successful cases. Balanced Accuracy equally considers sensitivity and specificity, making it more appropriate for clinical prediction tasks.

## Q8. What are the most important features in your model?

SHAP analysis showed that donor type, KDPI, cold ischemia time, donor-recipient distance, and recipient age were among the most influential features. These findings are also clinically consistent with existing transplant research.

## Q9. How does Federated Learning preserve privacy?

In Federated Learning, raw patient data never leaves the hospital. Only model parameters or weight updates are shared with the central aggregation server.

## Q10. Did you implement Differential Privacy or Secure Aggregation?

In this current study, we mainly focused on evaluating the baseline federated learning framework and optimization algorithms. Differential Privacy and secure aggregation are planned as future enhancements.

## Q11. What are the limitations of your work?

The study currently uses only 5 transplant centers and an MLP architecture. Future work will include larger federated networks, more advanced deep learning models, and stronger privacy-preserving mechanisms.

## Q12. Why is Explainable AI important in healthcare?

Healthcare professionals need to understand why a model makes a prediction before trusting it in clinical practice. SHAP provides transparency by showing which clinical features contributed most to the prediction.

## Q13. Could this system be deployed in real hospitals?

Yes, Federated Learning is highly suitable for real hospital environments because it allows collaboration without direct data sharing. Production deployment would require stronger security and clinical validation.

## Q14. How is your work different from existing transplant prediction studies?

Most existing transplant prediction studies use centralized datasets. Our work focuses on privacy-preserving collaborative learning across multiple transplant centers using Federated Learning.

## Q15. What is the main takeaway from your research?

Federated Learning can achieve performance close to centralized learning while preserving patient privacy, making it a promising solution for collaborative healthcare AI systems.

# Presentation Tips for Your First International Conference

- Speak slowly and confidently.
- Pause before answering questions.
- It is okay to say: “That’s an interesting question.”
- If you do not know an answer, say: “That is an interesting direction for future work.”
- Most researchers are discussing ideas, not attacking your work.
- You know your research better than anyone in the room.